

# Nathan Kwan-Ho Ng

Ph.D. Candidate, Computer Science  
University of Massachusetts Amherst

✉ kwanhong@cs.umass.edu

☎ (347) 615-8340

🌐 [linkedin.com/in/nathan-kwanho-ng](https://www.linkedin.com/in/nathan-kwanho-ng)  
🌐 [khngnathan.github.io/personal-website](https://khngnathan.github.io/personal-website)

## Education

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### Ph.D. in Computer Science

09/2019 – 12/2026 (Expected)

*University of Massachusetts Amherst*

Advisor: Prof. Prashant Shenoy

### B.Sc. in Computer Science

09/2014 – 06/2018

*Hong Kong Baptist University*

Advisor: Prof. Byron Choi

## Research Interests

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My research lies broadly in distributed systems and systems for machine learning, with a current emphasis on AI systems across the device–edge–cloud continuum. I design, build, and optimize systems and resource-management techniques that deliver predictable low-latency performance, covering both average-case and tail-latency behavior, for emerging AI applications.

## Publications

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### HYDRA: Multi-Level Ensemble Learning for Failure-Resilient Edge Inference

Krishna Praneet Gudipaty, Walid Hanafy, **Nathan Ng**, Kaan Ozkara, Qianlin Liang, Jesse Milzman, David Irwin, Suhas Diggavi, Prashant Shenoy

*ACM/IEEE Symposium on Edge Computing (SEC'26)*. To appear.

### Collaborative Processing for Multi-Tenant Inference on Memory-Constrained Edge TPUs

**Nathan Ng**, Walid Hanafy, Prashanthi Kadambi, Balachandra Sunil, Ayush Gupta, David Irwin, Yogesh Simmhan, Prashant Shenoy

*IEEE International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT'26)* **Best Paper Award**

### To Offload or Not To Offload: Model-driven Comparison of Edge-native and On-device Processing in the Era of Accelerators

**Nathan Ng**, David Irwin, Ananthram Swami, Don Towsley, Prashant Shenoy

*ACM/SPEC International Conference on Performance Engineering (ICPE'26)*

### TailClipper: Reducing Tail Response Time of Distributed Services Through System-Wide Scheduling

**Nathan Ng**, Abel Souza, David Irwin, Don Towsley, Prashant Shenoy

*ACM Symposium on Cloud Computing (SoCC'24)*

### Collaborative Inference in Resource-Constrained Edge Networks: Challenges and Opportunities

**Nathan Ng**, Abel Souza, Suhas Diggavi, Tarek Abdelzaher, Don Towsley, Prashant Shenoy

*IEEE Military Communications Conference (MILCOM'24), IoT for Adversarial Environments Workshop*

## Unlocking Efficiency: Understanding End-to-End Performance in Distributed Analytics Pipelines

Abel Souza, **Nathan Ng**, Tarek Abdelzaher, Don Towsley, Prashant Shenoy

*IEEE Military Communications Conference (MILCOM'23), IoT for Adversarial Environments Workshop*

## FGreat: Focused Graph Query Autocompletion

**Nathan Ng**, Peipei Yi, Zhiwei Zhang, Byron Choi, Sourav S. Bhowmick, Jianliang Xu

*IEEE International Conference on Data Engineering (ICDE'19), Demonstration Track*

## Answering the Why-Not Questions of Graph Query Autocompletion

Guozhong Li, **Nathan Ng**, Peipei Yi, Zhiwei Zhang, Byron Choi

*International Conference on Database Systems for Advanced Applications (DASFAA'18)*

## Preprints & Under Submission

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### Computation Placement in the Device-Edge Continuum: Model-Driven Comparison of On-Device, Edge-Native, and Split Processing with Accelerators

**Nathan Ng**, David Irwin, Ananthram Swami, Don Towsley, Prashant Shenoy

Under review at *ACM Transactions on Autonomous and Adaptive Systems (TAAS)*

### Tuning the Tail Latency of Distributed Queries Using Replication

**Nathan Ng**, Hung Le, Prashant Shenoy, Marco Serafini

Under review at the *Distributed and Parallel Databases Journal (DAPD)*. arXiv:2212.10387

## Patents

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### Dynamic user profile management

**Nathan Ng**, Tung Mai, Thomas Greger, Kelly Quinn Nicholes, Antonio Cuevas, Saayan Mitra, Somdeb Sarkhel, Anup Bandigadi Rao, Ryan A. Rossi, Viswanathan Swaminathan, Shivakumar Vaithyanathan

*US Patent US12,339,916B2*. Published June 2025.

### Dynamic user profile projection

**Nathan Ng**, Tung Mai, Thomas Greger, Kelly Quinn Nicholes, Antonio Cuevas, Saayan Mitra, Somdeb Sarkhel, Anup Bandigadi Rao, Ryan A. Rossi, Viswanathan Swaminathan, Shivakumar Vaithyanathan

*US Patent US12,130,876B2*. Published Oct 2024.

## Awards and Honors

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| <b>Best Paper Award</b> | 2026 |
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*DCOSS-IoT'26*

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| <b>Donald F. Towsley Graduate Scholarship in Computer Science</b> | 2026 |
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*University of Massachusetts Amherst*

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| <b>James Kurose Scholarship in Computer Science</b> | 2020 |
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*University of Massachusetts Amherst*

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| <b>Undergraduate Overseas Summer Research Scholarship</b> | 2018 |
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*Hong Kong Baptist University*

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| <b>Outstanding Student Scholarship</b> | 2017 |
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*Hong Kong Baptist University*

## Research Experience

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**Research Assistant, *University of Massachusetts Amherst*** 09/2022 – Present

Advisor: Prof. Prashant Shenoy

- Developed SwapLess, a system enabling multi-tenant collaborative inference on memory-constrained Edge TPUs.
- Developed an analytic-model-driven resource manager for adaptive inference offloading in edge environments.
- Developed TailClipper, a queueing-theory-inspired scheduler that reduces tail latency of distributed applications.

**Research Scientist Intern, *Adobe*** 06/2026 – 09/2026

Mentors: Dr. Haoliang Wang, Dr. Kunjal Panchal, Dr. Jack Wang

- Developing privacy-preserving LLM inference offloading strategies within Adobe Acrobat.

**Research Scientist Intern, *Adobe*** 05/2022 – 08/2022

Mentor: Dr. Tung Mai

- Developed an Observability Data Management System deployed in production for Adobe engineering teams.
- Developed a URL knowledge extraction system using Random Forest to improve metrics reporting.

**Data Science & Machine Learning Intern, *Adobe*** 05/2021 – 12/2021

Mentor: Dr. Tung Mai

- Proposed a cache admission algorithm using a predictive statistical model to forecast customer revisits.
- Proposed dynamic Cloud–Edge user profile replication algorithms to maximize cache hit rates.

**Student Research Collaborator, *Meta*** 01/2021 – 05/2021

Mentor: Dr. Ritwik Yadav

- Benchmarked performance of RocksDB and LiveGraph using LinkBench on AWS EC2 instances.
- Collaborated on a potential deployment of LiveGraph as a Database-as-a-Service model at Meta.

**Research Assistant, *University of Massachusetts Amherst*** 09/2019 – 05/2022

Advisor: Prof. Marco Serafini

- Proposed data replication algorithms to satisfy user-defined tail latency bounds for distributed queries.
- Developed a distributed graph query processing middleware that coordinated query execution across nodes.

**Research Assistant, *Hong Kong Baptist University*** 09/2018 – 08/2019

Advisor: Prof. Byron Choi

- Proposed algorithms for efficient indexed maximum common subgraph mining in graph databases.

**Undergraduate Student Researcher, *Hong Kong Baptist University*** 09/2017 – 08/2018

Advisor: Prof. Byron Choi

- Proposed ranking and search algorithms to address the ‘why-not’ questions in graph query autocompletion.

**Undergraduate Research Intern, *Nanyang Technological University*** 05/2017 – 08/2017

Advisor: Prof. Sourav S. Bhowmick

- Developed FGGreat, an interactive graph query autocompletion system that assists query formulation.

## Teaching Experience

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**CS 377: Operating Systems, Undergraduate Teaching Assistant** Spring 2025

*University of Massachusetts Amherst*

- Led lab sessions for 50 undergraduates covering OS concepts and hands-on assignments.

**CS 677: Distributed & Operating Systems, Graduate Teaching Assistant** Spring 2023, 2024

*University of Massachusetts Amherst*

- Held office hours, graded coursework, and provided Piazza support for students.

**CS 532: Systems for Data Science**, Graduate Teaching Assistant Fall 2019, 2022, 2023  
*University of Massachusetts Amherst*

- Designed and led lab sessions on Hadoop and Spark for 120 graduate students.

**CS 2007: Object-Oriented Programming**, Undergraduate Teaching Assistant Spring 2018  
*Hong Kong Baptist University*

- Assisted students with hands-on debugging and Java programming exercises.

**CS 1005: Essence of Computing**, Undergraduate Teaching Assistant Fall 2017  
*Hong Kong Baptist University*

- Assisted freshman students with programming exercises and debugging.

## Mentoring

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|--------------------------------------------------------------------|-------------|
| Sribatscha Maharana, B.Sc. and M.S. in Computer Science (UMass'27) | Spring 2026 |
| Naga Surya Prathik Vunnavu, M.S. in Computer Science (UMass'26)    | Spring 2026 |
| Balachandra Devarangadi Sunil, M.S. in Computer Science (UMass'26) | Fall 2025   |
| Ayush Gupta, M.S. in Computer Science (UMass'26)                   | Fall 2025   |
| Khai Nguyen, M.S. in Computer Science (UMass'25)                   | Spring 2025 |
| Tanay Joshi, M.S. in Computer Science (UMass'25)                   | Spring 2025 |
| Rishi Patel, B.Sc. in Computer Science (UMass'25)                  | Spring 2025 |

## Service & Outreach

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**External Reviewer** Fall 2025  
*ACM/IEEE Symposium on Edge Computing (SEC'25)*

**Shadow PC Member** Fall 2025  
*The European Conference on Computer Systems (EuroSys'26)*

**Instructor** Summer 2024, 2025  
*UMass Turing Program*

## References

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**Prashant Shenoy**  
 Distinguished Professor  
*University of Massachusetts Amherst*  
 shenoy@cs.umass.edu

**David Irwin**  
 Professor  
*University of Massachusetts Amherst*  
 irwin@ecs.umass.edu

**Hui Guan**  
 Assistant Professor  
*University of Massachusetts Amherst*  
 huiguan@cs.umass.edu

**Tung Mai**  
 Senior Research Scientist  
*Adobe*  
 tumai@adobe.com

**Don Towsley**  
 Distinguished Professor  
*University of Massachusetts Amherst*  
 towsley@cs.umass.edu